Production Planning and Control, aggregate production planning

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Production Planning and Control

The process of organizing production activities to achieve efficient resource utilization and meet delivery schedules.

Focus

- - Enhancement of productivity
- - Reduction of costs
- - Timely delivery

Objectives of PPC

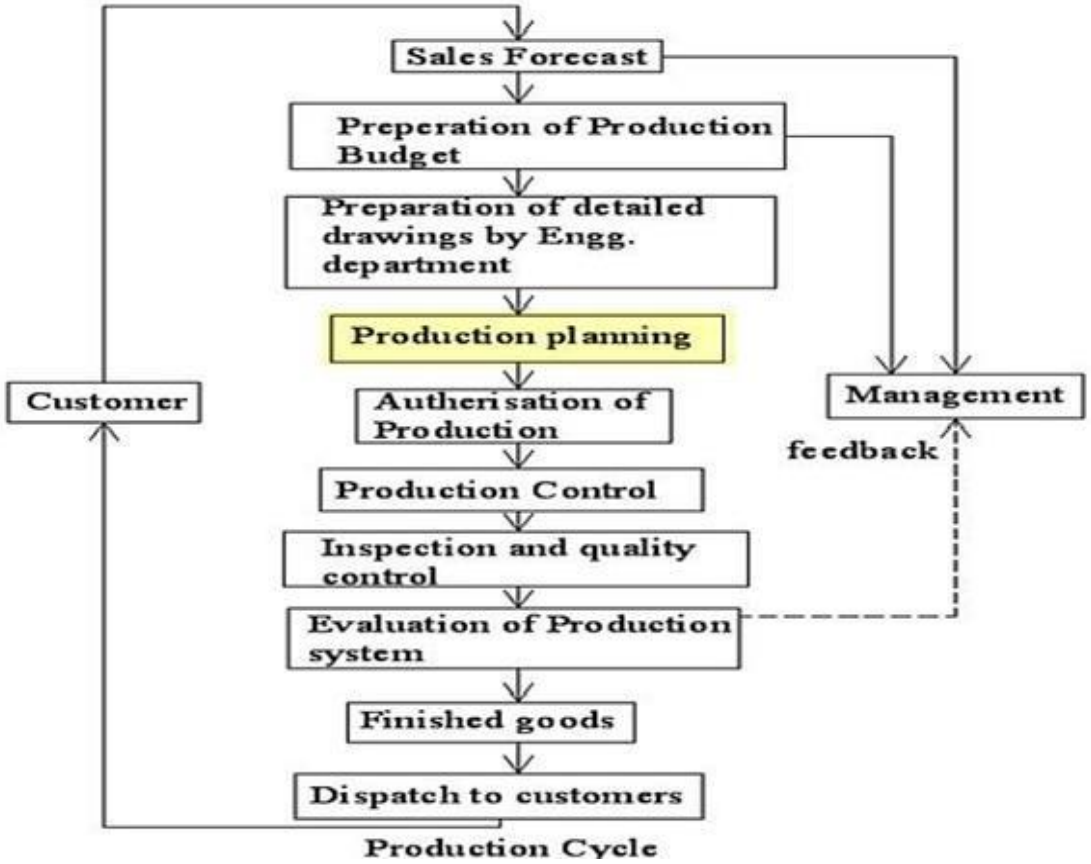
- - Efficient resource utilization
- - Meeting delivery schedules
- - Reducing costs
- - Improving product quality

Functions of PPC

- 1. Planning
- 2. Routing: Defining the sequence of operations and processes required to produce a product or service
- 3. Scheduling
- 4. Dispatching: Issuing instructions to start production activities and ensuring that they are carried out according to the schedule
- 5. Follow-up

PPC Cycle Overview

- Planning → Routing → Scheduling → Dispatching → Follow-up



Forecasting in PPC

Purpose

Predict future demand

Techniques

Qualitative: Expert judgment

Quantitative: Time series analysis

Capacity Planning

Ensuring production capacity meets demand.

Capacity planning involves determining the resources (like facilities, equipment, workforce, and materials) needed to meet current and future demand, ensuring a business can produce or deliver what customers need.

Methods

- Long-term: Facility layout
- Short-term: Resource allocation

Planning Process

Planning process can be

- **Long range plans (over 1 year):** Top executives are involved. Matters related to Research and Development, new product plans, capital investment and facility location/capacity are discussed.
- **Intermediate-range plans (3 to 12 months):** Operations managers with sales and operations planning teams are involved. Matters related to sales and operations, production planning and budgeting, employment and inventory are discussed.
- **Short-range plans (up to 3 months):** Operations manager with supervisors and foremen are involved. Issues of job assignments, ordering, scheduling, dispatching and overtime are discussed.

Sales and Operations (S&OP) Planning

- Intermediate planning is initiated by a process known as Sales and Operations Planning (S&OP**).**
- S&OP is a process of balancing resources and forecasted demand. It aligns an organization's competing demands from supply chain to final customer.
- One of the tasks of S&OP is to determine which plans are feasible in the coming months and which are not.
- When the resources appear to be at odds with market expectations, S&OP provides advanced warning to top management. If the plan cannot be supported in the long run, strategic changes need to be made.

Sales and Operations (S&OP) Planning

- The output of S&OP is called an aggregate planning. It is concerned with determining the quantity and timing of production for the intermediate future, often from 3 to 12 months ahead.
- Aggregate plans use information regarding product families or product lines rather than individual products. These plans are concerned with the total (aggregate) of the individual product lines.
- Companies have developed several systems for S&OP**,** each with own planning focus.
- For example, Company A may use S&OP with a focus on production decisions, Company B may use S&OP for inventory decisions and Company C may use S&OP with a focus on expensive investments.

Sales and Operations Planning (S&OP)

- An effective S&OP process needs some additional features to generate a useful aggregate plan:
  - A logical unit for measuring sales and output.
  - A forecast of demand for a reasonable intermediate planning period in aggregate terms.
  - A method to determine the relevant costs.
  - A model that combines forecasts and costs so that scheduling decisions can be made for the planning period.

Aggregate Planning Strategies

- The objective of aggregate planning is usually to meet forecast demand while minimizing cost over the planning period.
- When generating an aggregate plan, the operations manager must answer these questions:
  - Should inventories be used to adapt to the changes in demand during the planning?
  - Should changes be accommodated by varying the size of the workforce?
  - Should part-timers be used, or should overtime and idle time absorb fluctuations?
  - Should subcontractors be used on fluctuating orders?
  - Should prices or other factors be changed to influence demand?

Aggregate Planning Strategies

- **Capacity Options:** Basic capacity (production) options are
  - **Changing inventory levels:** Managers can increase inventory during periods of low demand to meet high demand in the future. If this strategy is selected, the cost of storage, insurance and handling will increase. On the other hand, low inventory will result in increasing lead time and poor customer service.
  - **Varying workforce size by hiring or layoffs:** A way to meet demand is to either hire or lay off production workers to match production rates. However, new employers need to be trained. Productivity drops temporarily as they are absorbed into the workforce. On the other hand, layoff lowers the enthusiasm and morale of the workers that lowers productivity.

Aggregate Planning Strategies

- **Capacity Options:**
  - **Varying production rates through overtime or idle time:** Keeping a constant workforce with varying working hours is possible. Whenever demand fluctuates towards the higher side, there is a limit on overtime. Overtime pay increases costs. However, too much overtime results in worker fatigue and a drop in productivity.
  - **Subcontracting:** A firm can acquire temporary capacity by subcontracting work during peak demand periods. However, subcontracting has several pitfalls. It may be costly. Also, it risks opening the door to a competitor.

Aggregate Planning Strategies

- **Demand Options:** Basic demand options are
  - **Influencing Demand:** When demand is low, a company attempts to increase demand through marketing, advertising, personal selling, and price cuts. However, sometimes these attempts are not able to balance demand with production productivity.
  - **Back ordering during high demand periods:** Back orders are orders for goods or services that a firm accepts but is unable to fill at the moment. If customers are willing to wait without loss of their goodwill or order, back ordering is a possible strategy. This approach often results in lost sales.

Revenue Management

- Most of the business models assume that firms charge all customers the same price for a product.
- In this planning category, the idea is to match capacity and demand by charging different prices based on the customer's willingness to pay.
- The challenge is to identify those differences and price accordingly. The technique for multiple price points is called **revenue management**.
- Also referred to as **yield management,** it determines the allocation of resources to maximize revenue or yield.
- For example, airlines alter the ticket fare based on demand, weather conditions and festive seasons.

Revenue Management

- To make revenue management work, the companies manage several issues:
  - ***Multiple pricing structures:*** These structures must be feasible and appear logical to the customer. These justifications may take various forms. For example, first-class seats and window seats on an airline.
  - ***Forecasts of the use and duration of the use:*** How many economy seats should be available? How much will customers pay for a hotel room with an ocean or mountain or scenic view?
  - ***Changes in demand:*** This means managing and dealing with issues that occur because the pricing structure may not seem logical to all customers. It means managing new issues, such as overbooking because the forecast was not perfect.

Material Requirements Planning (MRP)

Materials Requirements Planning (MRP) is a **systematic approach** used in manufacturing to **ensure the right materials are available at the right time in the right quantity** for production. It helps manage **inventory, production scheduling, and procurement** efficiently.

- Inputs:
- - Bill of Materials (BOM)
- - Inventory levels
- Outputs:
- - Material orders, production schedules

Dependent Inventory Model Requirements

- The requirements for the effective use of dependent inventory models are as follows:
  - Master production schedule (what is to be made and when)
  - Specifications or bill of material (materials and parts required to make the product)
  - Inventory availability (what is in stock)
  - Purchase orders outstanding (what is on order, also called expected receipts)
  - Lead times (how long it takes to get various components)

Master Production Schedule (MPS)

- MPS specifies what is to be made (the number of finished products or items) and when. The schedule (a timetable) must be in accordance with an aggregate plan.
- It includes a variety of inputs such as financial data, customer demand, engineering capabilities, labor availability, inventory fluctuations and supplier performance.
- As the planning process moves from the aggregate plan to execution, each of the lower-level plans must be feasible. When one is not, feedback to the next higher level is required to make the necessary adjustment.
- MPS addresses how to satisfy demand by specifying what items to make and when. While the aggregate plan is established in gross terms, the master production schedule is established in terms of specific products.

Master Production Schedule

- Master Production Schedule (MPS) is a statement of what is to be produced; it is *not a forecast*. The master schedule is expressed as follows:
  - A customer order in a job shop (make-to-order) company

    Example: print shops, machine shops, dining restaurants and Juice centre
  - Modules in a repetitive (assemble-to-order or forecast) company

    Example: customized cars and motorcycles, fast-food restaurant
  - An end item in a continuous (stock-to-forecast) company

    Example: steel, furniture items, electrical items such as light bulbs

Bill of Material

- A bill of material (BOM) is a list of quantities of components, ingredients, and materials required to make a product.
- List of information to include in BOM record:
  - **Part Name and Description:** The name of each part or assembly is recorded. This helps in identifying parts more easily. Also, a detailed description of each part helps in distinguishing between similar parts and identifying specific parts more easily.
  - **BOM Level:** Assign each part or assembly a number to detail where it fits in the hierarchy of the BOM. This allows anyone with an understanding of the BOM structure to quickly decipher the BOM.

Bill of Material

- **Procurement Type:** Documentation is to be done on how each part is purchased or made (i.e. off-the-shelf or made-to-specification) to create efficiencies in manufacturing, planning, and procurement activities.
- **Reference Designators:** If a product contains printed circuit board assemblies, you should include reference designators that detail where the part fits on the board in your BOM. Capturing this information in the BOM can save time and helps in avoiding confusion down the road.
- **BOM Notes:** Capture other relevant notes to keep everyone who interacts with BOM on the same page.

https://www.arenasolutions.com/resources/category/bom-management/creating-a-bill-of-materials/

Bill of Material

- **Phase:** Record what stage each part is in its lifecycle. For parts in production, it is common to use term like '*In Production*' to indicate the stage of the part. New parts that are not approved can be classified as '*Unreleased*'. This is helpful during new product introduction. It allows to easily track progress and create realistic project timelines.
- **Unit of Measure:** Classify the measurement in which a part will be used or purchased. Standard measures like inches, feet or ounces are suitable classifications. Consistency is to be maintained across all similar part types. This makes sure the right quantities are procured and delivered to the production line.

https://www.arenasolutions.com/resources/category/bom-management/creating-a-bill-of-materials/

Bill of Material

❑ Modular Bills

- It is used by those companies where the product has several optional features.
- These are bills of material organized by major subassemblies or by product options. They are major components of the final product or product options.
- Bills of material for modules are called modular bills. These bills are convenient because scheduling and production are facilitated by organizing modules rather than a multitude of final assemblies.

Example of Modular Bill:

- Say a washing machine is to be purchased. The available options are shown in Table. One option is to make a bill of every possible options (5×2×3×4=120 options).
- In such cases, modular bills are used.

Washing Machine			
Loading Capacity	**Loading type**	**Colour**	**Motor**
5 options	2 options	3 options	4 options

- The product is a washing machine for cleaning clothes. The available choices are are shown in Table. After adding the possible number of modular bill of material, the total number of bills is obtained as **1+5+2+3+4=15** for a total of **120 product combinations**

Component	Features
❑ Washing machine	1
➢ Loading capacity options	5
➢ Loading type options	2
➢ Colour options	3
➢ Motor options	4
Total modular bills	15
Total possible options	120

https://www.youtube.com/watch?v=8WIVY0mAS2M

Bill of Material

❑ Planning Bills

- Planning bills (also called pseudo or super bills) are created in order to assign an artificial parent to the bill of material. Such bills are used to group subassemblies so the number of items to be scheduled is reduced. Also, it is used to issue “kits” to the production department.
- For example, it may not be efficient to issue inexpensive items such as washers, nuts and cotter pins with each of the numerous subassemblies. Hence, “kits” are used to generate a planning bill. The planning bill specifies the kit to be issued. A planning bill may also be known as kitted material, or kit

Bill of Material

❑ Phantom Bills

- Phantom bills of material are bills of material for components that exist only temporarily. These components go directly into another assembly and are not inventoried.
- Hence, components of phantom bills of material are coded to receive special treatment. They are handled as an integral part of their parent item.
- An example is a transmission shaft with gears and bearings assembly that is placed directly into a transmission.

Lot-sizing decisions

- An MRP system is an efficient way to do production planning and determine net requirements. However, net requirements still demand a decision about how much and when to order. This decision is called **lot-sizing decision**. Some techniques of lot-sizing decisions are
  - **Periodic Order Quantity:** Periodic order quantity (POQ) is a lot-sizing technique that orders the quantity needed during a predetermined time between orders, say every 4 weeks. POQ is the order quantity that covers the specific demand for that interval.

Lot-sizing decisions

- **Lot-for-Lot:** When frequent orders are economical (when setup costs are low) and just-in-time inventory techniques are implemented, lot-for-lot is used. However, when setup costs are significant, lot-for-lot can be expensive.
- **Economic Order Quantity (EOQ):** EOQ is useful for cases with relatively constant demand. EOQ lot sizing often does not perform well in MRP with varying. Sometimes, operations managers take advantage of demand information when it is known. In such cases, EOQ is used to do lot sizing for comparison purposes.

Closed-Loop MRP

Closed-loop material requirements planning implies an MRP system that provides feedback to scheduling from the inventory control system. A closed-loop MRP system provides information to the capacity plan, master production schedule, and ultimately to the production plan.

Manufacturing Resource Planning (MRP II)

Manufacturing Resource Planning (MRP II) is an **advanced version of Materials Requirement Planning (MRP)** that integrates **all aspects of manufacturing**, including **production planning, scheduling, inventory control, finance, and workforce management**.

It provides a **holistic approach** to managing **resources, materials, and business operations** to improve efficiency and decision-making.

Enterprise Resource Planning (ERP)

- Definition:
- Integrated system for managing business processes.
- Role in PPC:
- - Real-time data sharing
- - Streamlined operations

Aggregate Planning

- Aggregate planning is a mid-term planning process (3–18 months) that determines the quantity and timing of production to meet demand, while minimizing costs by optimizing resource allocation and managing fluctuations in demand.
- Objectives: Optimize production, inventory, and workforce
- Strategies: Level, chase, and hybrid approaches
- Level capacity planning strategy involves maintaining a stable production rate regardless of the fluctuations in demand.
- Chase capacity planning strategy involves adjusting the production rate in line with the fluctuations in demand.
- Hybrid capacity planning strategy is a combination of level and chase capacity planning. This strategy involves maintaining a stable production rate for the core products while using chase capacity planning for the products with fluctuating demand.

Production Scheduling

Gantt Chart

Work	0-9 months	9-18 months	18-27months	27-36months
Planning for bonding mechanism (theoretical study)				
Collection of materials for preparation specimen				
Procurement of AAC blocks with and without bonding mechanism				
Experimental evaluation of AAC blocks wall system without any bonding mechanism				
Experimental evaluation of AAC blocks wall system with bonding mechanism				
Preparation of report and guidelines				

The **Master Production Schedule (MPS)** is a **detailed plan** that outlines **what to produce, how much to produce, and when to produce** it over a specific period.

Inventory Control

- Types:
- - Raw materials, WIP, finished goods
- Techniques:
- - Economic Order Quantity (EOQ)
- - ABC Analysis

Follow-up and Feedback

Purpose

- Monitoring progress, identifying deviations, and implementing corrective actions.
- KPIs (key performance indicators): Delivery times, resource utilization.

Quality Control in PPC

- Integration: Total Quality Management (TQM).
- Purpose: Maintain product consistency and meet standards.

Challenges in PPC

- - Demand fluctuations
- - Supply chain disruptions
- - Resource constraints.

Technologies in PPC

Examples:

- - Artificial Intelligence (AI)
- - Internet of Things (IoT)
- - Digital Twins

Future Trends in PPC

- - AI-driven planning
- - Data analytics
- - Automation and robotics